



DEPARTMENT OF
COMPUTER SCIENCE



DATA TYPES, VARIABLES, AND OPERATORS

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LEARNING OBJECTIVES

At the end of the lesson, you should be able to:

- Explain the role of programming languages in computer programming.
- Discuss how problems are represented as data.
- Apply data types, variables and operators to problem solving.



PROGRAMMING AND PROGRAMMING LANGUAGES

A programme is a series of instructions that causes a computer or a microcontroller to perform a particular task.

A computer program includes more than just instructions. It also contains data and various memory addresses in which the instructions work to perform a specific task.



PROGRAMMING AND PROGRAMMING LANGUAGES

Examples of Programming languages are C, C++, Rust, Assembly, Java, Python, etc.
Let see some of their syntaxes





PROBLEMS AND DATA /VARIABLES

When we programme problems, we represent the problem as data (or variable) and the variable is stored and processed by the computer.



PROBLEMS AND DATA/VARIABLES

Example: To find the average of the first 5 multiples of 9, what are the variables?





DATA TYPES

Data type is used for specifying the type of a variable. They determine the type and size of data associated with variables. There are three basic types in the C language: characters, integer, and floating-point numbers.



DATA TYPES

There are three basic types in the C language: characters, and integer and floating-point numbers.

char	usually 8-bits (1 byte)
int	usually the natural word size for a machine or OS (e.g., 16, 32, 64 bits)
short int	at least 16-bits
long int	at least 32-bits
float	usually 32-bits
double	usually 64-bits
long double	usually at least 64-bits



DATA TYPES

DATA TYPE	MEMORY SIZE (BYTES)	RANGE
signed char	1	-128 to 127
unsigned char	1	0 to 255
short int	2	-32,768 to 32,767
unsigned short int	2	0 to 65,535
int	4	-2,147,483,648 to 2,147,483,647
unsigned int	4	0 to 4,294,967,295
long	8	-9223372036854775808 to 9223372036854775807
unsigned long	8	0 to 18446744073709551615
long long int	8	-9223372036854775808 to 9223372036854775807
unsigned long long int	8	0 to 18446744073709551615



VARIABLE DEFINITION AND DECLARATION

Variable definition always includes “data type” followed by “variable name”.

A variable definition is also a declaration, but not all variable declarations are definitions. What does this mean?

Data type Variable name

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unsigned char cityXTemperature;

cityXTemperature = 25;



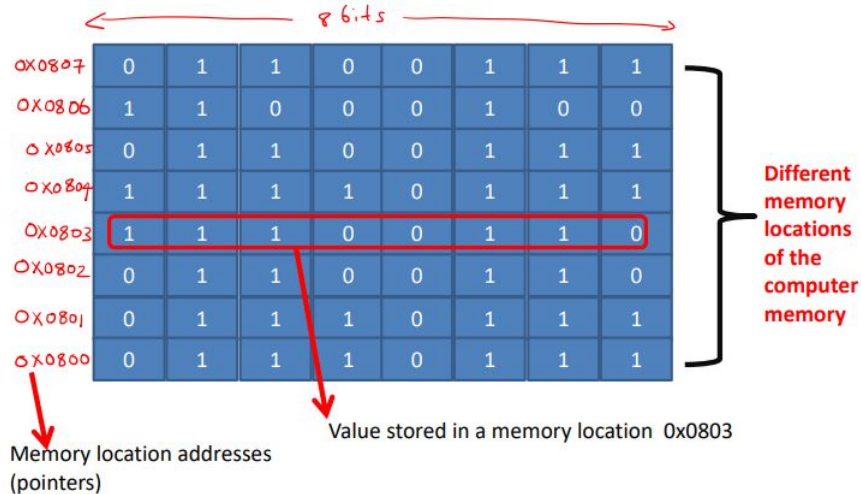
VARIABLE DEFINITION AND DECLARATION

Before using a variable, you should always define it by using appropriate data type. You can select the data type according your programme logic and need.



THE COMPUTER MEMORY

Variable names are not stored inside the computer memory; the compiler replaces them with memory location addresses.





OPERATORS

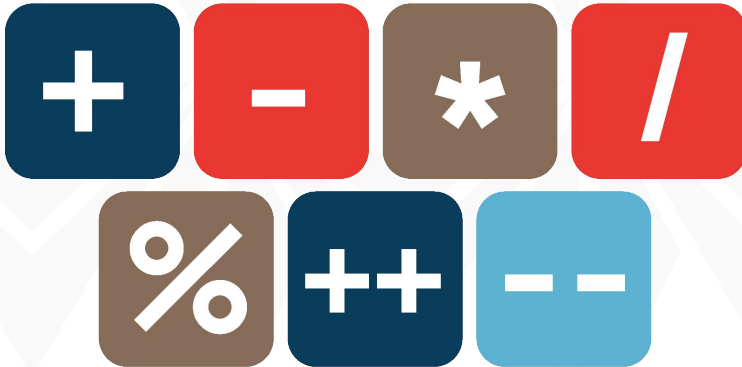
An operator is a symbol that tells the compiler to perform specific mathematical or logical functions. The C language is rich in built-in operators and provides the following types of operators:

- Arithmetic operators
- Relational operators
- Logical operators
- Bitwise operators
- Assignment operators
- Misc operators



ARITHMETIC OPERATORS

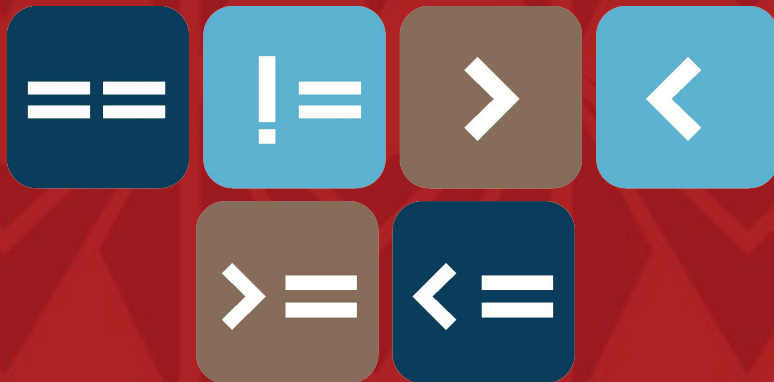
Examples of arithmetic operators





RELATIONAL OPERATORS

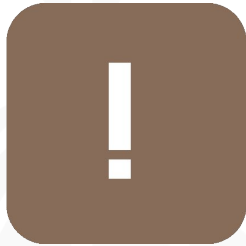
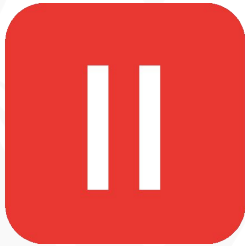
Examples of relational operators





LOGICAL OPERATORS

Examples of logical operators





ASSIGNMENT OPERATORS

Examples of assignment operators

$=$

$+=$

$-=$

$*=$

$/=$

$\%=$



MISC OPERATORS

Examples of misc operators

`sizeof()`

`&`

`*`

`?:`



SUMMARY

- A programme is a series of instructions that causes a computer or a microcontroller to perform a particular task.
- Variable definition always includes “data type” followed by “variable name”.
- Variable names are not stored inside the computer memory; the compiler replaces them with memory location addresses.
- Data type is used for specifying the type of a variable.



FURTHER READING RESOURCES

- Bailey, T. (2005). An introduction to the c programming language and software design.
- Kernighan, B. W., & Ritchie, D. M. (2002). The C programming language.



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THANK
YOU